

electric charge - practice problems

name _____

1. Two plastic strips are rubbed with the same wool cloth on a dry winter day, and both strips gain electrons. Do the strips attract, repel, or have no electric force on each other?

2. On a dry winter day, a metal doorknob loses 2.5×10^{10} electrons to your hand. What is the doorknob's charge?

3. A balloon rubbed on your hair ends up with a charge of -20 nC. How many extra electrons does it have?

4. A tiny oil droplet in a charge experiment has a charge of $+4.80 \times 10^{-18}$ C. How many electrons is the droplet missing?

5. A typical lightning bolt carries about 15 C of negative charge from a cloud to the ground. How many electrons is that?

6. A balloon with a charge of -32 nC touches a neutral metal sphere, and some of the balloon's charge moves onto the sphere. What charge does the sphere end up with?
7. In 1909, Robert Millikan measured the charges on tiny oil droplets. Suppose four droplets have charges of -3.2 , -4.8 , -8.0 , and $-11.2 \times 10^{-19} \text{ C}$. Show that each charge is a whole number of electrons. How many extra electrons does each droplet have?
8. A neutral sweater and a neutral balloon are rubbed together. The sweater loses 5.0×10^{11} electrons to the balloon. What is the final charge on each?
9. A comb is charged by running it through dry hair. It then picks up tiny bits of paper, even though the paper is neutral. Why?
10. A small bead starts with a charge of $+2.40 \times 10^{-18} \text{ C}$. After rubbing, its charge is $-8.00 \times 10^{-19} \text{ C}$. How many electrons did the bead gain or lose?
11. A Van de Graaff generator's metal dome builds up a charge of $+10 \text{ } \mu\text{C}$. How many electrons have been removed from the dome?